C++ Notes: Working With Strings

(Summary from the Strings section of the course)

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Learning Objectives

After this section, you will be able to:

- Print strings using `cout`
- Use `endl` and `\n`
- Store strings inside variables
- Use `length()`, `find()`, and `substr()`
- Access and modify characters using indexes

Core Idea

Core idea

A string represents plain text in C++. We can measure it, search inside it, modify it, and extract parts of it.

Printing Strings

```
cout << "Draft Academy";
```

Output:

```
Draft Academy
```

Using endl

```
cout << "Draft Academy" << endl;
cout << "Hello";
```

Output:

```
Draft Academy
Hello
```

Without endl:

```
cout << "Draft Academy";
cout << "Hello";
```

Output:

```
Draft AcademyHello
```

Using \n

```
cout << "Draft Academy\nHello";
```

Output:

```
Draft Academy
Hello
```

Storing Strings in Variables

Before:

```
string phrase = "Draft Academy";
cout << phrase;
```

Output:

```
Draft Academy
```

length()

```
string phrase = "Draft Academy";
cout << phrase.length();
```

Output:

```
15
```

Accessing Characters

```
string phrase = "Draft Academy";
cout << phrase[0];
```

Output:

```
D
```

Indexing starts at 0.

Modifying Characters

Before:

```
Draft Academy
```

After:

```
string phrase = "Draft Academy";
phrase[0] = 'B';
cout << phrase;
```

Output:

```
Braft Academy
```

find()

```
string phrase = "Draft Academy";
cout << phrase.find("Academy", 0);
```

Output:

```
6
```

substr()

```
string phrase = "Draft Academy";
cout << phrase.substr(6, 3);
```

Output:

```
Aca
```

Storing substring:

```
string phrase = "Draft Academy";
string part = phrase.substr(6, 3);
cout << part;
```

Output:

```
Aca
```

Common Mistakes

Common mistakes

- Forgetting indexing starts at 0
- Mixing single and double quotes
- Using an index beyond string length

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